Project #15 | Fall 2026 | Lehman College
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Public GitHub repository:
https://github.com/boatenghelvin-crypto/Event-Volunteer-Management-System

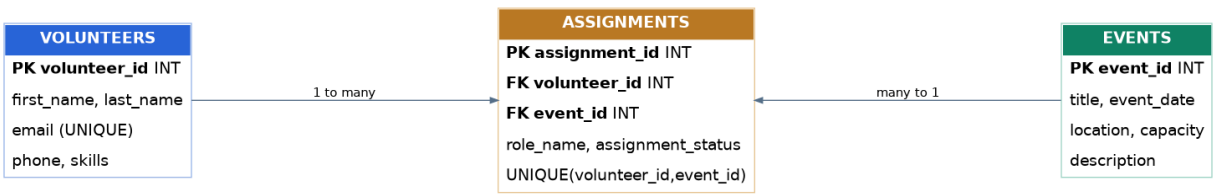
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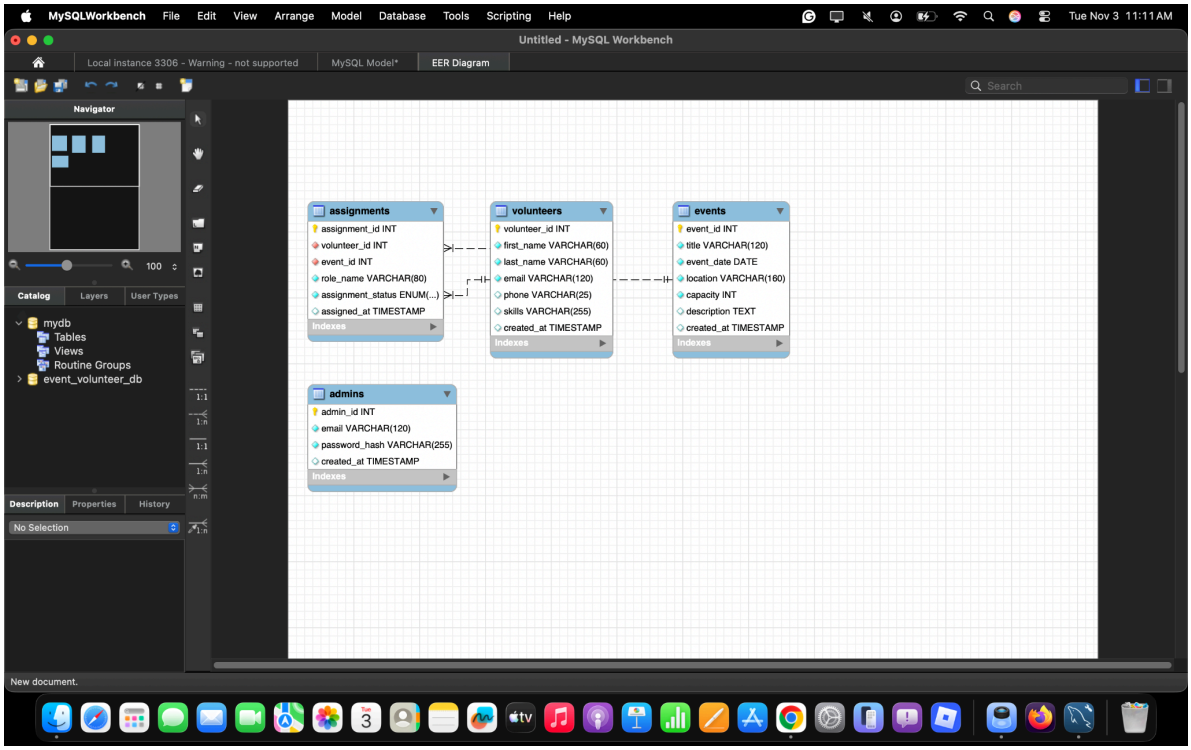
1. Project overview
VolunteerHub is an online application for managing community volunteer activities, events, and event schedules. It allows administrators to create volunteer profiles, register activities, and assign volunteers to events. This system prevents the occurrence of duplicate events and exceeding the maximum allowed number of volunteers.

2. Project plan and scope
• **Goal**: Replace disconnected volunteer lists and sign-up sheets with one searchable relational database.
• **Users**: Authorized community event coordinators.
• **Front end**: Responsive HTML and CSS with basic JavaScript confirmations.
• **Back end**: PHP 8 and MySQL, accessed through PHP PDO.
• **Deliverables**: Functional application, database, SQL join examples, GitHub repository, report, user guide, and demo recording

3. Main features
• Administrator setup and login with hashed passwords.
• Create, edit, and delete volunteer records, including contact details and skills.
• Create, edit, and delete events, with date, location, and maximum volunteer capacity.
• Assign volunteers to events, specify roles, and update the status as Registered, Confirmed, Completed, or Cancelled.
• Display summary metrics, capacity counts, and recent assignments on the dashboard.

4. Database design
The database has four tables, which include the admins table, volunteers table, events table, and assignments table. The assignments table solves the many-to-many relationship that the volunteers have with the events. Each assignment table contains two foreign keys, where a combination of the volunteer_id and the event_id produces a unique assignment.





5. Table descriptions

Table	Main columns	Purpose
volunteers	volunteer_id (PK), first_name, last_name, email, phone, skills	Stores contact and skills.
events	event_id (PK), title, event_date, location, capacity, description	Stores event details.
assignments	assignment_id (PK), volunteer_id (FK), event_id (FK), role_name, assignment_status	Connects volunteers and events.
admins	admin_id (PK), email, password_hash	Authenticates coordinators.

6. SQL joins for data retrieval/Code Snippets

The dashboard demonstrates both INNER JOIN and LEFT JOIN, an important grading requirement.

INNER JOIN (three tables): Retrieve volunteer names, event names, roles, and statuses. The output only includes assignments linked to valid volunteer and event records.

```
SELECT v.first_name, v.last_name, e.title, a.role_name

FROM assignments a
JOIN volunteers v ON a.volunteer_id = v.volunteer_id
JOIN events e ON a.event_id = e.event_id;
```

LEFT JOIN: Show every event and count its active assignments, including events with no assignments.

```
SELECT e.title, COUNT(a.assignment_id) AS total
FROM events e LEFT JOIN assignments a
ON e.event_id = a.event_id AND a.assignment_status <> 'Cancelled'
GROUP BY e.event_id, e.title;
```

7. Code and processing workflow

To log in, the coordinator selects a page of the application and completes the relevant form. The PHP technology checks the name and value of the request and the CSRF token. PDO interacts with MySQL using a prepared statement. The process results in success or failure and redirects the user to the relevant page.

9.Conclusion

VolunteerHub demonstrates the relationship between events and volunteers through a relational database. SQL joins help the system connect the data in various tables and provide administrators with beneficial information. It grants administrators the power to manage volunteers, organize events, allocate volunteers to events, and trace their participation in them. Thanks to PHP and MySQL, VolunteerHub makes data storage and retrieval convenient and efficient. Therefore, this project shows the benefits of a relational database for successful event management and avoiding mistakes at the same time.